In the case of a wireless signal, the property that varies (such as voltage in a wired signal) is the electric field. Electric fields do not require a material medium to exist or vary in. According to Maxwell's equations, an alternating (varying) electric field gives rise to a similarly varying magnetic field, and hence such a signal is called an electromagnetic wave. Such waves travel far and wide in space be it vacuum, or occupied with matter. The varying electric field gives rise to a varying magnetic field, which in turn, gives rise to an electric field, and so on. The basic

Band name	Abbr	Frequency and wavelength in air	Uses
Extremely low frequency	ELF	3-30 Hz 1,00,000 km - 10,000 km	Communication with submarines
Super low frequency	SLF	30-300 Hz 10,000 km - 1,000 km	Communication with submarines
Ultra low frequency	ULF	300-3000 Hz 1,000 km - 100 km	Communication within mines
Very Low frequency	VLF	3-30 kHz 100 km - 10 km	Submarine communication, avalanche beacons, wireless heart rate monitors, geophysics
Low frequency	LF	30-300 kHz 10 km - 1 km	Navigation, time signals, AM longwave, broadcasting, RFID
Medium frequency	MF	300-3000 kHz 1 km - 100 m	AM (Medium-wave) broadcasts
High frequency	HF	3-30 MHz 100 m - 10 m	Shortwave broadcasts, amateur radio and over-the-horizon aviation communications, RFID
Very High frequency	VHF	30-300 MHz 10 m - 1 m	FM, television broadcasts and line-of-sight ground-to-aircraft and aircraft-to-aircraft communications. Land Mobile and Maritime Mobile communications
Ultra High frequency	UHF	300-3000 MHz 1 m - 100 mm	television broadcasts, microwave ovens, mobile phones, wireless LAN, Bluetooth, GPS and Two-Way Radios such as Land Mobile, FRS and GMRS Radios
Super High frequency	SHF	3-30 GHz 100 mm - 10 mm	microwave devices, wireless LAN, most modern Radars

The various frequencies and their usage.